MODULE 6

OSINT: Open Source Intelligence

Certified Cyber Criminologist - Police / LEA Training

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What You Will Learn

By the end of this module you will turn one email, one number or one username into a **verified intelligence picture** - legally.

- **What OSINT is** - and how it differs from ordinary Googling
- **Investigating from** an email, a phone number, a username - step by step
- **Google dorking** - advanced search commands with safe examples
- **Email headers, reverse image search, WHOIS, photo metadata** - every lead has a tool
- **Sock puppets and OPSEC** - investigating without being seen

The Case That Opens This Chapter

An investigator has **only one email ID**. The fraudster sent a fake job offer from it. No name, no number, no face.

- Three hours later he has the fraudster's **name, social accounts, old website, and two associates**
- No magic happened. What happened is **OSINT** - Open Source Intelligence
- The art of **methodically connecting** information available in the open
- In four labs you will try it yourself - on your own data first

What is OSINT?

OSINT is intelligence from **open sources** - information available to everyone on the internet.

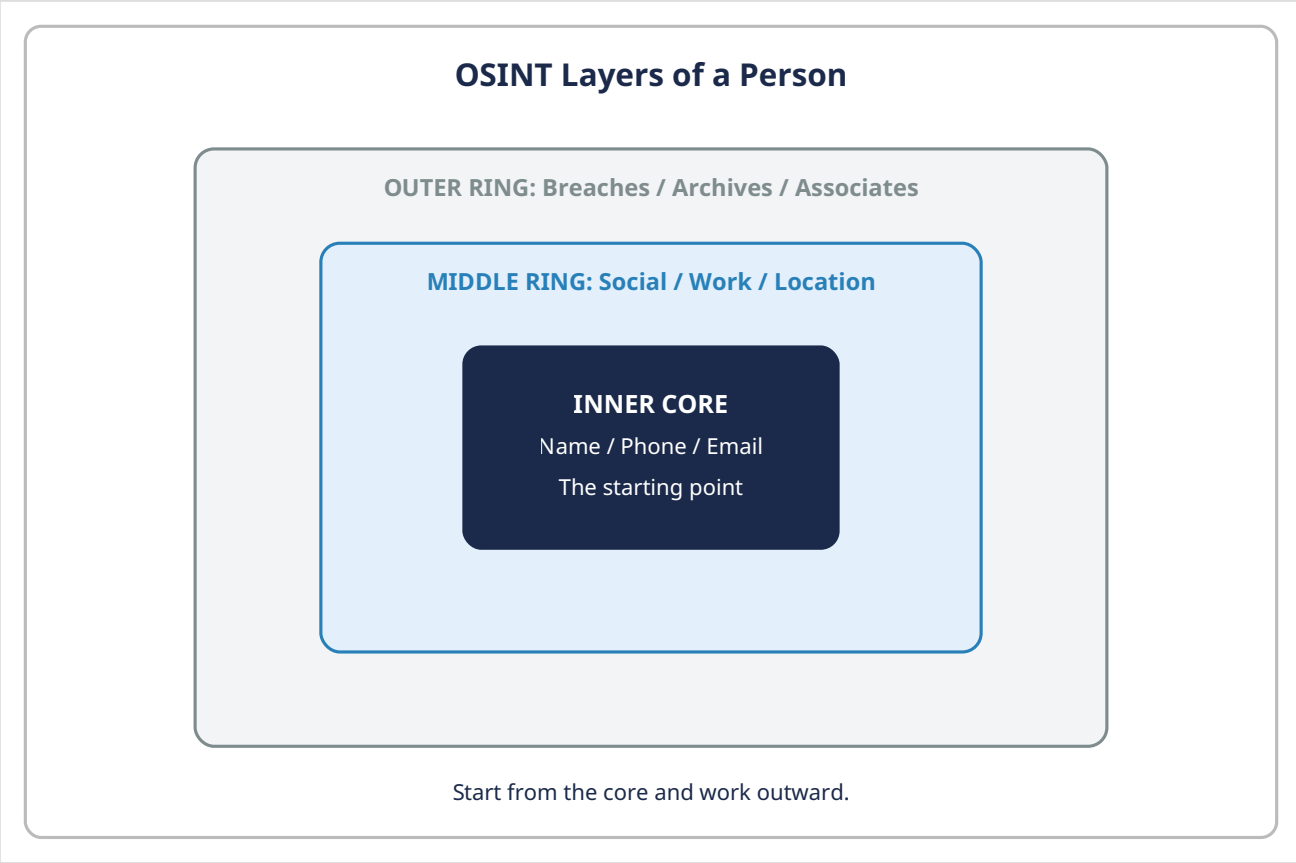
- Social media, news, public records, websites, forums - **all public, all legal to view**
- You already do it: Googling a name, checking a Facebook profile. The investigator does it **methodically**
- Methodical means: **note everything, connect every link, verify every claim**
- People leave a lot behind in public - that is OSINT's power and its responsibility

What OSINT Can and Cannot Do - Legally

OSINT is powerful, but it has **hard legal walls**. Know them before you start.

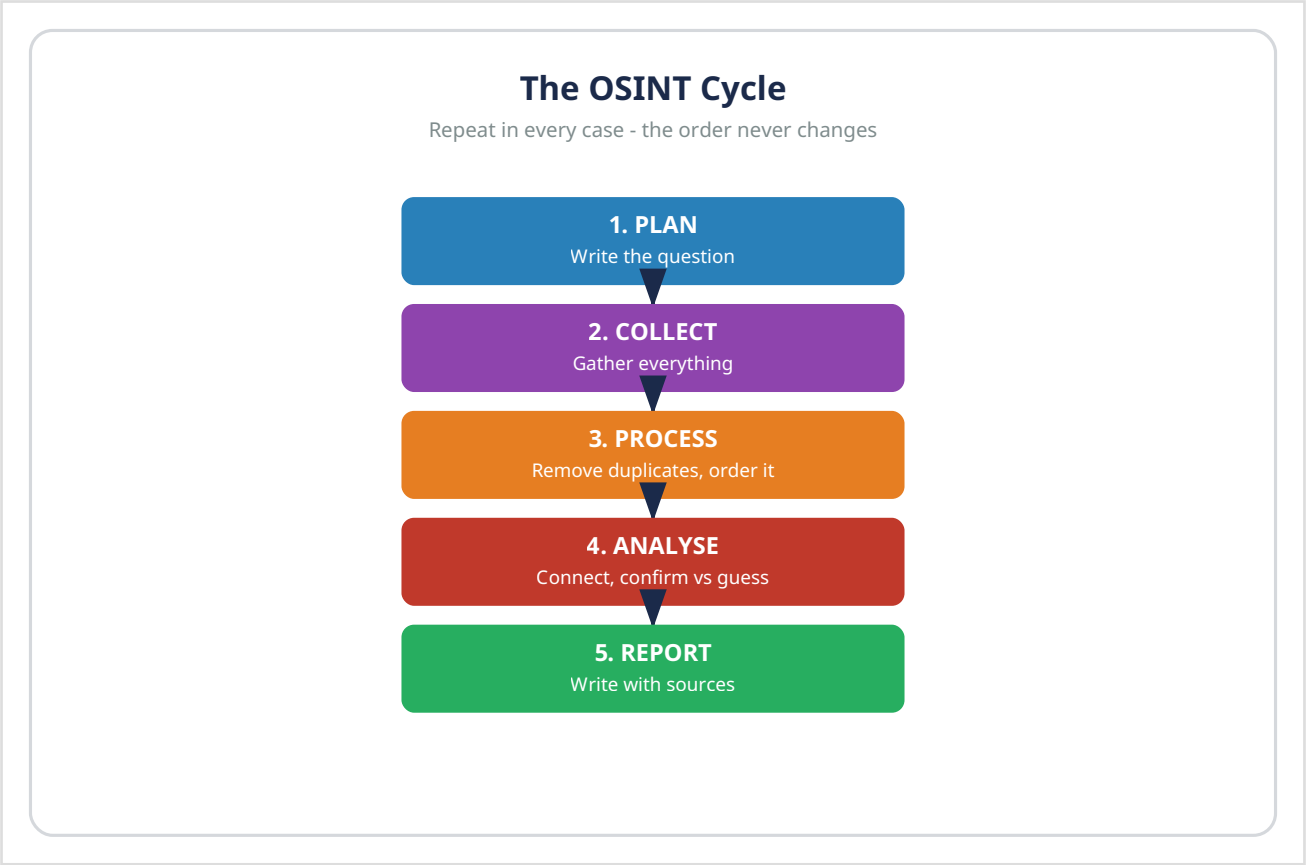
- **CAN:** view anything public - profiles, posts, breach data, archived pages
- **CANNOT:** hack accounts, intercept communications, or impersonate a real person - that is crime
- Private information needs **legal process** - BNSS notices, not clever tricks
- The **Puttaswamy** privacy line applies to you too - excitement never justifies overreach

OSINT Layers of a Person



- Start at the **inner core** - name, phone, email - then move outward
- Each layer - social, work, location, breaches, associates - **feeds the next**
- Never skip layers: the **outer rings** often hold the breakthrough

The OSINT Cycle



- **OSINT without order is wandering** - plan, collect, process, analyse, report
- Collect **without judging**; judge only in the analysis step
- Report **what was NOT found** too - that is part of the truth

Investigating from an Email

One email ID can open an entire case. **Four routes, in order.**

- **HaveIBeenPwned:** has this email appeared in a data breach? Old leaks reveal linked data
- **Email header:** 'Show original' in Gmail - the **Received** lines expose the real sender's IP
- **Google in quotes:** 'thug123@gmail.com' - find where else this email was used
- **Social signup check:** many people register social accounts with the same email

Reading an Email Header

The header is the part users never see - **for you it is treasure**. It holds the email's complete journey.

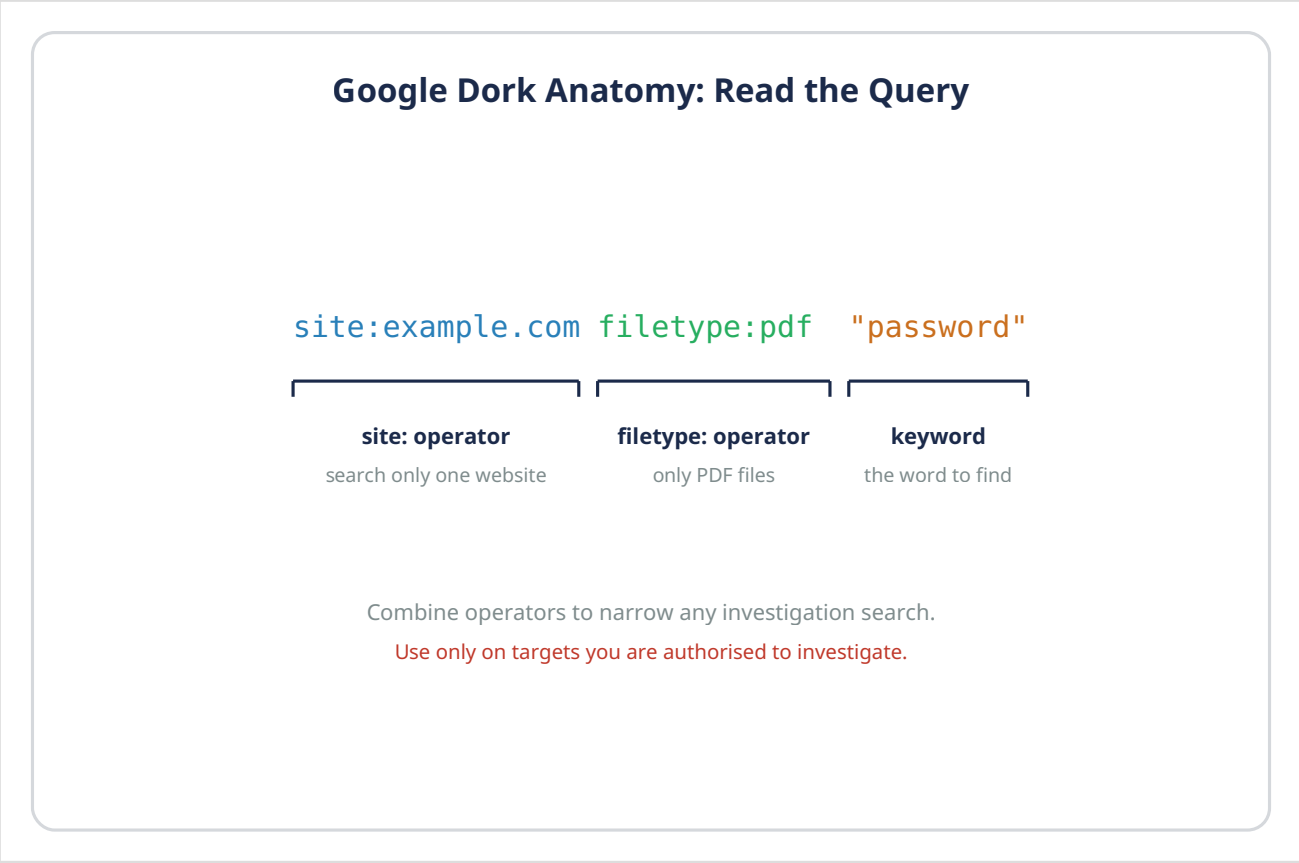
- Gmail: three dots, **'Show original'** - the full technical record appears
- **Received lines** read bottom-to-top; the bottom line is closest to the real sender
- **From vs Reply-To vs Return-Path** mismatch means possible spoofing - be alert
- **SPF, DKIM, DMARC** all failing means the email is likely fake; the IP reveals country and company

Investigating from a Phone Number

In fraud cases you often start with **just one number**. Here is the ladder.

- **Truecaller-type apps:** a name guess - an **initial lead only, never evidence**
- **WhatsApp check:** profile photo, status, online timing - from a **sock puppet**, never your real ID
- **Google the number** in different formats - it often sits on ads, websites or complaint boards
- The starting digits reveal the **telecom circle** - a small fact that gives direction

Google Dorking: Advanced Search



- Read the query **left to right**: operator, operator, keyword
- Dorking finds what **ordinary search never shows**
- Combine commands - precision beats luck

Dorking Cheat Sheet



- **site:** searches inside one website; **filetype:** finds only PDFs or docs
- **Quotes** force exact matches; **intitle: / inurl:** hunt login and admin pages
- Practice on **your own** name first - Lab 2 is built for this

Dorking: The Legal Line

Dorking is legal - **what you do with the findings decides everything**.

- Viewing **public** information is legal; **misusing** a discovered weakness is a crime
- Try dorks only on **your own information** - never on someone else's name or email
- Safe example: **site:linkedin.com 'cyber security' india** - public, useful, lawful
- An investigator **gathers** information; he never exploits it

Investigating from a Username

People reuse the **same username everywhere** - one name can unlock ten accounts.

- **Namechk / WhatsMyName:** search one username across hundreds of sites at once
- Try **variants**: cooldude123, cooldude1234, cool_dude123 - people keep similar names
- **Old forgotten accounts** are goldmines - a ten-year-old profile may hold the real photo or city
- Note the **date and source** of every account found - a profile without a source is useless

Reverse Image Search

Fraudsters steal faces. **Reverse search steals them back** - it tells you where a photo really comes from.

- **Google Lens / TinEye:** upload the photo or paste its link; see every public copy
- A 'job recruiter's' photo turned out to be a **foreign model** - case cracked on the spot
- Works in **matrimonial fraud** too - stolen photos are the standard trick
- Limit: public photos only - and a match is a **lead, not evidence**, until joined with more

WHOIS and the Wayback Machine

Every fake website leaves **two public trails** - its birth record and its old faces.

- **WHOIS:** when was the domain created? A 'bank' site born **two weeks ago is fake**
- **SSL certificate:** click the browser's lock icon - if the name and site differ, be cautious
- **Wayback Machine** (web.archive.org): the site's old form - a **third party's record**, usable in court
- Note every snapshot's **URL and date** - that is your evidence chain

Photo Metadata (EXIF)

A digital photo carries **more than the picture** - camera, date-time, sometimes the exact location.

- **ExifTool** (free) extracts the complete hidden metadata of a photo
- Social media **strips** most metadata on upload - but **email attachments** keep it
- A criminal's photo can betray his **device and location** - check before you dismiss it
- Flip side: **strip metadata** before sharing your own photos - good OPSEC habit

Sock Puppet: 5 Safety Rules



- View the criminal from your **real ID** and he will **delete evidence and run**
- Five rules: separate profile, separate email, no real photo, VPN on, **view only**
- The sock puppet is a **tool**, not a disguise for mischief

Sock Puppet: The Full Rules

A fake identity for investigation is allowed - **inside strict walls**. Cross them and the tool becomes a crime.

- **View only:** never talk to anyone, provoke anyone, or lay traps from it
- Never wear a **real person's** face - no stolen photos, no real names; use a neutral avatar
- **Record every account:** where created, which case - and stop using it when the case ends
- Zero link to your real identity: separate browser profile, separate email, **no personal overlap**

The Investigator's OPSEC

While you watch the criminal, **make sure he cannot watch you back**.

- **Separate everything:** device or browser profile for investigation - real Gmail and banking never inside
- **Location services off** during investigation; VPN on to mask your IP
- Notes and screenshots where **only authorised people** can reach
- **Silence on social media:** one small post about an ongoing case can destroy it

Six Mistakes That Ruin OSINT (Part 1)

In OSINT, **mistakes damage more than missing tools**. The first three:

- **1. Real-ID investigation:** the suspect sees you coming, deletes evidence, vanishes
- **2. One-source truth:** a name on one website is not the truth - verify from **two separate sources**
- **3. Guess becomes fact:** 'seems from Indore' and 'is from Indore' go in **separate columns**
- Write it wrong once and your whole report's credibility cracks

Six Mistakes That Ruin OSINT (Part 2)

And the remaining three - **all about discipline, not skill**:

- **4. No source written:** found it but didn't note where - now it can neither be verified nor used in court
- **5. Old as new:** a five-year-old profile is not today's truth - **date everything**
- **6. Crossing the privacy line:** extracting private data without authority can put **you** in the dock
- Discipline beats tools - every time

360-Degree Profiling

Fragments become a person when you **join them** - name, phones, emails, accounts, job, friends, habits.

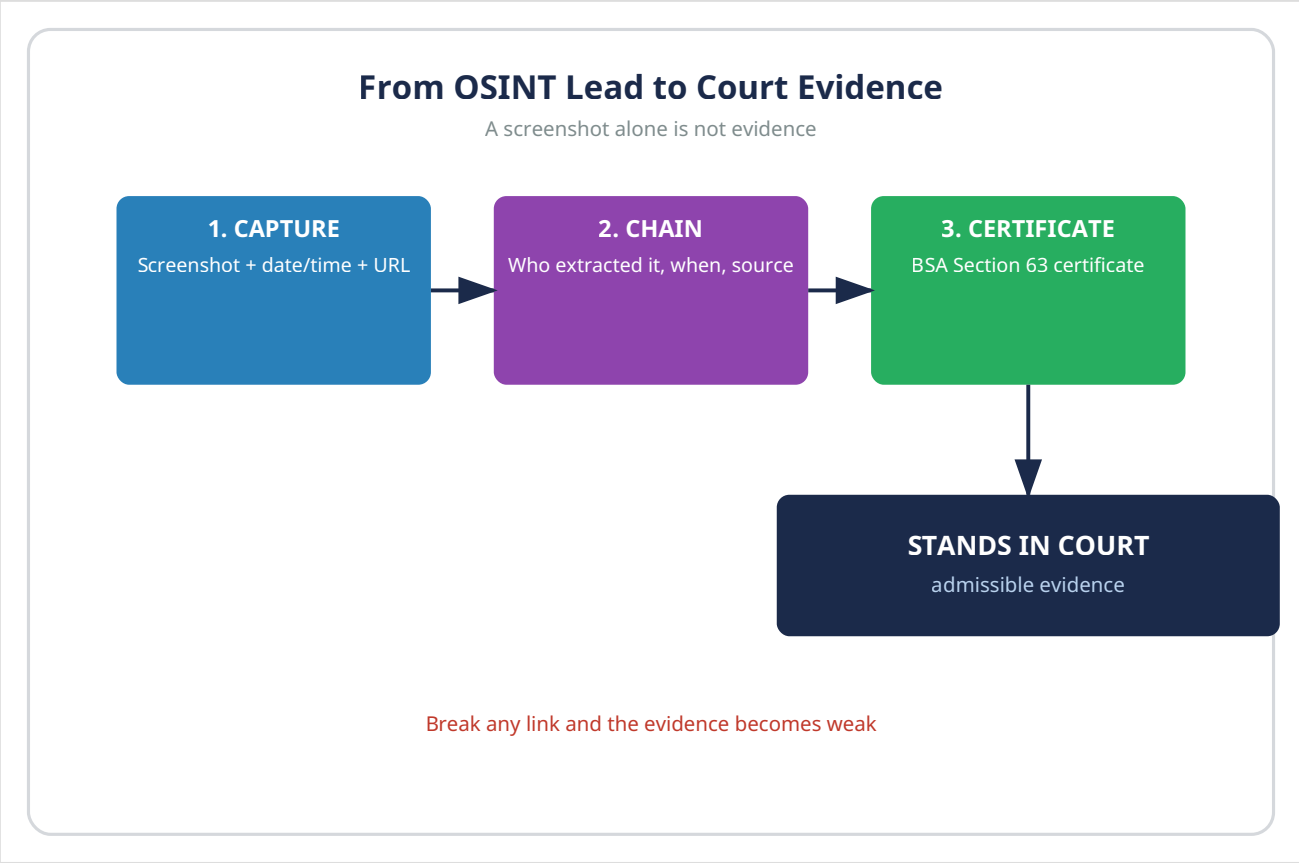
- Grade every fragment's source: **strong** (direct evidence), **medium** (inferable), **weak** (assumption)
- **Never fill empty boxes by guessing** - write what is NOT known, and the next lawful step
- Profiling serves **investigation, not harassment** - misusing personal data is a crime
- Real-person profiling needs **legal authority and a senior's permission** - Lab 4 uses a fictional person

Writing the OSINT Report

A bad report **kills a good investigation**. Structure yours like this:

- **Five parts:** case ID and purpose, methods used, results **with sources**, analysis, recommendation
- **Source with every claim** - information without a source is rumour
- Keep **confirmed facts and guesses** in separate sections - mixing them misleads
- Screenshots named methodically: **case12_facebook_profile_2026-10-14.png**, date-time visible

From OSINT Lead to Court Evidence



- A lead becomes evidence through **capture, chain, and certificate** - all three
- Screenshot + PDF save + **Wayback save** - so deletion cannot erase your proof
- Then the **BSA Section 63 certificate** - you learned its format in Module 5

Case Walkthrough: One Email, Full Investigation

A **fictional** practice case - watch every technique join into one method:

- **Step 1 - the email:** Google it in quotes, check HaveIBeenPwned, read the header, extract the IP
- **Step 2 - linked accounts:** run the username through Namechk; hunt the variants
- **Step 3 - social media:** note public photos, friends, locations - with dates and sources
- **Step 4 - website and report:** WHOIS the linked domain, Wayback its history, write it all up

Real Case: From One Number to the Gang

Many fraud cases began with **a single phone number** - and ended with a busted network.

- A name guessed on **Truecaller**, photo and status on **WhatsApp**, old complaints on **Google** - links joined
- The accused identified; then **CDR through legal process** exposed his entire network
- Lesson: a small lead gives a big result **when connected methodically**

Source: The Times of India

Real Case: Bulli Bai - Digital Investigation

2022 - women's photos misused through an app. The accused sat in **different states**.

- Mumbai Police followed **social accounts, IP addresses and digital footprints** across state lines
- Arrests followed - **distance meant nothing** against digital evidence
- Lesson: **OSINT plus digital forensics** solves even interstate crime

Source: The Indian Express

Your Turn: Labs in This Module

Four labs - **all on your own data or fictional persons only**. Never on someone else's private information.

- **Lab 1:** your own exposure report from HaveIBeenPwned - then change leaked passwords
- **Lab 2:** map your public footprint with Google dorking - then fix what should not be public
- **Lab 3:** build your sock-puppet safety checklist - all seven ticks before you investigate
- **Lab 4:** 360-degree profiling worksheet on a **fictional** person - grade every source

Key Takeaways

If you remember nothing else from Module 6, remember this:

- OSINT is **fully legal and fully methodical** - random searching is not OSINT
- Follow the **cycle**: plan, collect, process, analyse, report
- Email, phone, username, photo, domain - **every lead has a tool**
- **Sock puppet always** - your real ID never touches a suspect
- Verify from **two sources**; date everything; source everything
- A lead becomes evidence only via **capture + chain + BSA 63 certificate**

Review Questions

- 1. You have only a fraudster's email ID. List your first five investigative steps in order - and the legal boundary of each.
- 2. A Truecaller name says 'Rahul Sharma'. Can you name him as the accused in your report? Explain the correct handling.
- 3. Write three Google dorks you could safely run on your own information, and state what each reveals.
- 4. Your sock puppet gets a friend request from the suspect's associate. Do you accept? Justify with the rules.
- 5. Distinguish 'confirmed', 'medium' and 'weak' source information, with one example of each from the profiling lab.

NEXT: MODULE 7

Advanced Intelligence & Evidence Mapping

Standalone environment, SOCMINT, evidence maps, timelines